

Amar Prakash Pandey

Senior Data Engineer

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Summary

Senior Data Engineer with 8+ years of experience designing scalable, high-performance data architectures, ELT/ETL pipelines, and large-scale batch processing systems. Proficient in handling petabyte-scale datasets with Apache Spark, Databricks, Airflow, and dbt, backed by strong SQL, Python, and Scala. Skilled in building cloud-native data solutions on AWS and GCP, with a focus on performance optimisation, cost efficiency, and data quality.

Contributor to Apache Airflow; past maintainer at CRI-O and DuckDuckGo. Writes about Spark internals and performance tuning at amarpandey.me/blog

Experience

Citi Pune, India
Senior Data Engineer June 2025 – Present

- Working with the **Market Risk team** to analyse daily trade data and calculate key risk metrics across Citi's global trading operations.
- Building and optimising **big data pipelines** using **Spark, HDFS, and Autosys** to process and validate large volumes of trade data efficiently, using **GitHub Copilot with Claude models** to accelerate development and code review.
- Supporting risk reporting and analytics by ensuring high data quality, scalability, and timely delivery of risk calculations.

Sahaj Software Pune, India
Senior Data Engineer March 2021 - June 2025

Developed and maintained end-to-end batch pipelines to process audience and frame-level data for out-of-home (OOH) advertising, enabling accurate and timely delivery of insights to campaign optimisation and analytics teams. Owned data quality checks and pipeline performance end-to-end for effective ad targeting and impact measurement.

- **Designed and optimised ETL pipelines with Apache Spark (PySpark) on Databricks**, processing terabytes of data and executing large-scale joins to enrich advertisement analytics.
- **Enhanced Spark job performance by 50%** through advanced shuffle partition tuning and resource optimisation, significantly reducing processing time.
- **Implemented and managed** workflow orchestration with **Apache Airflow**, ensuring efficient sequencing, automated retries, and SLA compliance for batch processing.
- **Cut S3 storage costs by 60%** by implementing efficient data lifecycle policies and optimising file formats for cost-effective cloud storage.
- Automated deployment of **Airflow DAGs, Apache Spark/PySpark jobs, and microservices** using **GitLab CI/CD**, built with **Cursor AI** to deliver the setup in a fraction of the estimated time with broader test and edge-case coverage, eliminating manual errors and ensuring seamless promotion across dev and production environments.

Tata Consultancy Services Chennai, India
System Engineer December 2017 - February 2021

Developed a **pricing platform** that streamlined and automated pricing updates for thousands of components. My contributions focused on **optimizing workflows, improving system reliability, and enhancing deployment efficiency**.

- **Developed APIs** for Cummins' pricing platform using **Java, Spring Boot, and MySQL**, enabling real-time pricing updates for thousands of components.
- **Automated pricing workflows**, reducing update time from **one day to near real-time**.
- **Migrated Oracle trigger and function logic** to a service layer for **better logging and maintainability**.
- Automated deployment using CI/CD pipelines with Git, **Jenkins, and GitLab**, improving efficiency and reliability.

Skills

Programming Languages Java, Python, Scala, SQL 	Big Data Technologies Apache Spark, PySpark, Databricks, Apache Hadoop 	ETL Tools & Orchestration Apache Airflow, dbt (Data Build Tool) 
Cloud Technologies AWS S3 EMR, ECS, ECR, Lambda, RDS, Google Cloud Platform BigQuery 	Databases PostgreSQL, Redshift, DynamoDB, MongoDB, Oracle 	DevOps Git, Docker, Terraform, Jenkins, GitLab CI/CD 

Projects

<u>spot-optimizer</u>	January 2025
Built and published an open-source Python library on PyPI that replaces in-house logic for AWS Spot Instance selection and recommends the most stable, cost-effective instances based on cores, memory, region, and EMR version compatibility .	
<u>geohashviz.com</u>	July 2024
GeohashViz is a web-based visualization tool that helps users explore geohashes interactively. It allows users to encode locations into geohashes, visualize different precision levels, and understand spatial hierarchies	
<u>Finger Detection and Tracking</u>	April 2017
A computer vision project (400+ GitHub stars) that detects and tracks fingers in real time using OpenCV and Python , enabling gesture-based interactions from a live camera feed.	

Awards

Senior Data Engineer	March 2024
Recognized for exceptional contributions at Sahaj.ai , receiving multiple commendations for assisting projects, building useful tools, and consistently going above and beyond.	
System Engineer Cummins	June 2019
Awarded Star Performer of the Year at TCS for outstanding performance on the Cummins project.	
<u>Software Developer</u> Google Summer of Code	August 2017
Completed Google Summer of Code (2017) , contributing to the Teammates project at the National University of Singapore .	